

Chapter 8:

Integration Techniques and Improper Integrals

Section 8.7: Integration by Tables and Other Integration Techniques (Only Rational Functions of Since and Cosine)

Rational Functions of Sine and Cosine

$$\int \frac{1}{2 - 3\sin\theta}\, d\theta$$

$$\int \frac{1}{1 + \sin\theta + \cos\theta}\, d\theta$$

Substitution for Rational Functions of Sine and Cosine

For integrals involving rational functions of sine and cosine, the substitution

$$u = \frac{\sin x}{1 + \cos x} = \tan\frac{x}{2}$$

yields

$$\cos x = \frac{1 - u^2}{1 + u^2}, \quad \sin x = \frac{2u}{1 + u^2}, \quad \text{and} \quad dx = \frac{2\,du}{1 + u^2}.$$



Proof From the substitution for u, it follows that

$$u^2 = \frac{\sin^2 x}{(1 + \cos x)^2} = \frac{1 - \cos^2 x}{(1 + \cos x)^2} = \frac{1 - \cos x}{1 + \cos x}.$$

Solving for $\cos x$ produces

$$\cos x = \frac{1 - u^2}{1 + u^2}.$$

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To find $\sin x$, write $u = (\sin x)/(1 + \cos x)$ as

$$\sin x = u(1 + \cos x) = u\left(1 + \frac{1 - u^2}{1 + u^2}\right) = \frac{2u}{1 + u^2}.$$

Finally, to find dx, consider $u = \tan(x/2)$. Then you have $\arctan u = x/2$ and

$$dx = \frac{2\,du}{1 + u^2}.$$

EXAMPLE

Find $\int \frac{1}{1+\sin\theta+\cos\theta}\,d\theta$

SOLUTION

$$I = \ln\left|\tan\left(\frac{\theta}{2}\right) + 1\right| + C$$

Can We Integrate All Continuous Functions?

The answer is No, at least not in terms of the functions that we are familiar with

The functions that we have been dealing with in this book are called **elementary functions**. These are the polynomials, rational functions, power functions (x^n), exponential functions (b^x), logarithmic functions, trigonometric and inverse trigonometric functions, hyperbolic and inverse hyperbolic functions, and all functions that can be obtained from these by the five operations of addition, subtraction, multiplication, division, and composition. For instance, the function

$$f(x) = \sqrt{\frac{x^2 - 1}{x^3 + 2x - 1}} + \ln(\cosh x) - xe^{\sin 2x}$$

is an elementary function.

Can We Integrate All Continuous Functions?

The answer is No, at least not in terms of the functions that we are familiar with.

If f is an elementary function, then f' is an elementary function but $\int f(x)\,dx$ need not be an elementary function. Consider $f(x) = e^{x^2}$. Since f is continuous, its integral exists, and if we define the function F by

$$F(x) = \int_0^x e^{t^2}\,dt$$

then we know from Part 1 of the Fundamental Theorem of Calculus that

$$F'(x) = e^{x^2}$$

Can We Integrate All Continuous Functions?

The answer is No, at least not in terms of the functions that we are familiar with.

Thus $f(x) = e^{x^2}$ has an antiderivative F, but it has been proved that F is not an elementary function. This means that no matter how hard we try, we will never succeed in evaluating $\int e^{x^2}\,dx$ in terms of the functions we know. (In Chapter 11, however, we will see how to express $\int e^{x^2}\,dx$ as an infinite series.) The same can be said of the following integrals:

$$\int \frac{e^x}{x}\,dx \qquad \int \sin(x^2)\,dx \qquad \int \cos(e^x)\,dx$$

$$\int \sqrt{x^3 + 1}\,dx \qquad \int \frac{1}{\ln x}\,dx \qquad \int \frac{\sin x}{x}\,dx$$

In fact, the majority of elementary functions don't have elementary antiderivatives. You may be assured, though, that the integrals in the following exercises are all elementary functions.

Solve exercises 55-62 of section 8.7.

Start with the suggested problems mentioned in the syllabus.